

A multisectoral general equilibrium model of Schumpeterian growth and fluctuations

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Abstract

This paper builds a dynamic multisectoral general equilibrium model of Schumpeterian growth and fluctuations based on the endogenous introduction of new products. The set of technological opportunities determines the degree of product- quality upgrading and consists of breakthroughs and improvements the latter exhibiting diminishing returns. If the degree of diminishing returns to technological improvements is low, then there is no steady-state equilibrium but cycles of breakthroughs and improvements. We analyze two symmetric stable patterns of innovation which generate endogenous cycles and growth. There is a negative correlation between the duration of each cycle and the long-run growth trend.

Key words: Schumpeter; Endogenous growth; Endogenous cycles; Technological change; New products

JEL classification: E3; O3; O4

1. Introduction

Schumpeterian growth is a type of endogenous growth which is based on Schumpeter's (1942) description of innovation through creative destruction.

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Dinopoulos (1994) provides a recent overview of Schumpeterian growth theory.¹ The present paper generalizes the model of Schumpeterian growth developed by Segerstrom, Anant, and Dinopoulos (1990) by introducing a distinction between technological *breakthroughs* and *improvements*. This distinction results in endogenous deterministic cycles within a multisectoral dynamic general equilibrium framework. Segerstrom et al. focused on a steady-state equilibrium consisting of technological breakthroughs only.

The theoretical distinction between technological breakthroughs and improvements follows the spirit of Jovanovic and Rob (1990), Jovanovic and MacDonald (1994), and Bresnahan and Trajtenberg (1992). The first study analyzed the implications of extensive and intensive searches for production techniques, the second one focused on the effects of a technological breakthrough on the number of firms in an industry, and the third study examined the role of general purpose technologies (as opposed to specific purpose ones) on growth. The present paper complements these partial-equilibrium studies by concentrating on intersectoral innovation patterns consisting of technological breakthroughs and improvements which occur within a general equilibrium framework.

Examples of technological breakthroughs are major discoveries like the computer, the telephone, the television, the first antibiotic drug, etc. The development of personal computers, various versions of telephones (digital, cellular), color and high definition TVs, and various antibiotic drugs correspond to quality improvements upon the initial products.²

The model generates several interesting results. If the degree of diminishing returns to technological improvements is high (i.e., improvements are costly relative to breakthroughs), then there exist a unique steady-state equilibrium which consists of breakthroughs only, and there are no cycles. However, if the degree of diminishing returns to technological improvements is low (i.e., some improvements are less costly than breakthroughs), then a steady-state equilibrium does not exist, and the stable patterns of innovation generate endogenous deterministic cycles. Two symmetric cyclical patterns of innovation are analyzed: The first consists of a breakthrough–improvement sequence that shifts continuously across sectors. The second consists of a round of breakthroughs in every sector followed by rounds of improvements in every sector. Both cyclical patterns of innovations exhibit a sort of negative ‘duration dependence’ within

¹See also Cheng and Dinopoulos (1992b) and in particular Aghion and Saint-Paul (1993) for surveys of studies on Schumpeterian growth and fluctuations.

²See Bresnahan and Trajtenberg (1992) for an excellent description of the development and implications of process technological breakthroughs like the rotary motion and electronics. Jovanovic and MacDonald (1994) list specific major innovations in the automobile industry which were followed by improvements. Gort and Klepper (1982) present statistical evidence for 46 new products which supports the technological breakthrough–improvement hypothesis.

a sector and a negative correlation across sectors at a point in time. Finally, the long-run growth trend is inversely related to the length of a typical cycle for small changes in the parameters of the model. For example, a marginal increase in the labor endowment of the economy, or in the duration of the patent length results in a higher long-run growth trend and shorter cycles.

Section 2 of the paper develops the model. Section 3 analyzes the dynamic symmetric patterns of innovation and growth, and Section 4 concludes. The Appendix contains the proofs of propositions.

2. A model of growth through technological breakthroughs and improvements

Labor is the only factor of production, it is supplied inelastically, and can be allocated to two activities: manufacturing of final goods and production of R&D services. Workers can move freely across the two activities, and one unit of labor produces one unit of output in both manufacturing and R&D services. The wage rate is the numeraire, and it is set equal to unity.

Consumers supply labor, invest in firms, and consume goods. Firms maximize instantaneous profits in the product markets and net expected discounted profits when they invest in R&D. They are price takers except when they are winners of R&D races, in which case they earn monopoly profits until their technology becomes public knowledge. There is free entry in every R&D race. R&D races are modeled as invention lotteries, so that while individual firms face uncertainty there is no aggregate risk. At each instant in time, assets are equal to the cumulative value of net investment in R&D adjusted for capital gains and losses. The cumulative interest factor clears the stock market at each instant in time. The stock market channels consumer savings to firms engaged in R&D.

2.1. Consumer behavior

The intertemporal utility function of the representative consumer is given by

$$u = \int_0^{\infty} e^{-\rho t} \ln U(\cdot) dt, \quad (1)$$

where $\rho > 0$ is a constant subjective discount rate and $U(\cdot)$ is an instantaneous utility function to be defined in (3) below.

A product ‘generation’ is represented by the following sequence of potential product quality increments which are treated as parameters:

$$\hat{\alpha}_k = \frac{1}{(1 - \beta \mu^{-k})} \quad \text{for } k = 0, 1, 2, \dots, \infty. \quad (2)$$

Each $\hat{\alpha}_k$ corresponds to a potential product, and represents the extent of quality increment of a product over its immediate predecessor, in a sense that a unit of product k is equivalent to $\hat{\alpha}_k$ units of product $(k - 1)$. Parameter $0 < \beta < 1$ captures the quality level of each product, and parameter $\mu > 1$ captures the degree of diminishing returns across quality improvements. Each product generation starts with a breakthrough $\hat{\alpha}_0 = 1/(1 - \beta) > 1$ which can be followed by a sequence of diminishing improvements $\hat{\alpha}_k$ ($k > 0$). Each subsequent version can be discovered only after the previous version of a product has been discovered with $\hat{\alpha}_\infty \rightarrow 1$. Breakthroughs ‘replenish’ the set of technological opportunities in the sense that when a new breakthrough occurs after k improvements have been discovered, the improvement opportunity set $\{\hat{\alpha}_{k+1}, \hat{\alpha}_{k+2}, \dots\}$ is enlarged to $\{\hat{\alpha}_1, \hat{\alpha}_2, \dots\}$. There are n sectors in the economy, each of which consists of a countably infinite set of product generations.

The following notation is used throughout the paper. The presence of a hat over a quality increment denotes a *potential* quality increment, whereas the absence of a hat denotes the actual (or realized) quality increment. Superscript i in parenthesis denotes the sector ($i = 1, 2, \dots, n$). In addition, we use subscript j to keep track of sequential economy-wide R&D races. Finally, subscript s denotes the s th improvement of a product. For example, $\alpha_{sj}^{(i)}$ denotes the quality increment of the s th product in sector i resulting from race j . The value of $\alpha_{sj}^{(i)}$ is given by $\hat{\alpha}_k$ for some k ($k = 0, 1, \dots$) defined in (2). The use of this notation will become clear as we proceed.

The instantaneous utility function of a representative consumer is

$$U(x_0^{(1)}, x_1^{(1)}, x_2^{(1)}, \dots, x_0^{(2)}, x_1^{(2)}, \dots) = \prod_{i=1}^n \left[\sum_{q=1}^{\infty} \left[x_q^{(i)} \prod_{k=1}^q \alpha_k^{(i)} \right] + x_0^{(i)} \right], \quad (3)$$

where $x_q^{(i)}$ is the quantity of the q th product actually consumed in sector i and $\prod_{k=1}^q \alpha_k^{(i)}$ is the cumulative quality index of product q in the sector. The subscript j is omitted because it has no bearing on consumer utility. The parameter $\alpha_k^{(i)}$ is an element of (2) and reflects the extent of quality improvement of product k over product $(k - 1)$ in sector i . The sequence $\{\alpha_1^{(i)}, \alpha_2^{(i)}, \dots, \alpha_k^{(i)}, \dots\}$ is a sequence of product quality increments which have been actually discovered through R&D races in sector i , and would therefore enter the utility function of the consumer.³

³For example, if the fifth product developed in sector i is the immediate result of the first improvement which is preceded by two breakthroughs and two improvements, then $x_5^{(i)}$ enters the utility function as $\alpha_1^{(i)} \alpha_2^{(i)} \alpha_3^{(i)} \alpha_4^{(i)} \alpha_5^{(i)} x_5^{(i)} = \hat{\alpha}_0 \hat{\alpha}_1 \hat{\alpha}_2 \hat{\alpha}_0 \hat{\alpha}_1 x_5^{(i)}$ whereas product $x_4^{(i)}$ enters the utility function as $\alpha_1^{(i)} \alpha_2^{(i)} \alpha_3^{(i)} \alpha_4^{(i)} x_4^{(i)} = \hat{\alpha}_0 \hat{\alpha}_1 \hat{\alpha}_2 \hat{\alpha}_0 x_4^{(i)}$. Notice that each unit of product $x_5^{(i)}$ is equivalent to $\alpha_5^{(i)} = \hat{\alpha}_1$ units of product $x_4^{(i)}$.

Under Eq. (3), goods within a sector adjusted for quality are identical, whereas products across sectors are ‘grossly’ independent. The instantaneous utility function gives rise to a simple product replacement mechanism which is standard in all Schumpeterian growth models. First, to maximize (3) it suffices to consume only one of the available versions in each sector. Second, the price of one version constraints the price of the next one. For example, if $p_k^{(i)}$ is the price of product $x_k^{(i)}$, then

$$p_k^{(i)} \leq p_{k-1}^{(i)} \alpha_k^{(i)}. \quad (4)$$

If condition (4) fails, then the k th product cannot compete against the $(k-1)$ th product, and consequently will never be developed. The sole producer of a new product sets its price so that (4) is barely satisfied (i.e., as an equality) and that demand shifts completely to the state-of-the-art quality product in each sector. Once imitation occurs, perfect competition drives the price of a product down to its unit costs.

The Appendix derives the solution to the intertemporal consumer maximization problem which determines the optimal aggregate consumption expenditure, $E(t)$:

$$E^*(t) = \frac{r_c(t)}{\lambda_0 e^{\rho t}} = \frac{1}{\lambda(t)}, \quad (5)$$

where $\lambda_0 > 0$ is the marginal utility of wealth at $t = 0$, $\lambda(t)$ is the marginal utility of wealth at t , and $r_c(t)$ is the cumulative interest factor. If $r_c(t)$ is differentiable, then $r(t) = (dr_c(t)/dt)/r_c(t)$ is the instantaneous interest rate. In this case, differentiating $E^*(t)$ with respect to time, one could obtain the standard optimality condition $\dot{E}/E = r(t) - \rho$. The expression for $E^*(t)$ in (5) implies that consumption shifts to the future if the interest factor is higher than the impatience factor $e^{\rho t}$. In addition, if $r_c(t)$ jumps, there is a corresponding jump in $E^*(t)$ reflecting the consumer's optimal arbitrage.⁴

2.2. Product markets

We assume that the winner of each R&D race enjoys temporary monopoly power for a fixed period of time, T , which can be thought of as the effective patent length. Let $\alpha_k^{(i)}$ denote the quality increment associated with the state-of-the-art product k in sector i . The monopolist charges a price equal to $\alpha_k^{(i)}$ for a period of T . Condition (4) implies that all firms that know how to produce

⁴Cheng and Dinopoulos (1992a) analyze the consumer's arbitrage problem formally.

product $k - 1$ cannot make any positive profits and drop out of the market. The instantaneous profits of a monopolist in sector i producing product version k are given by

$$\Pi_k^{(i)} = (\alpha_k^{(i)} - 1) \frac{E(t)}{n\alpha_k^{(i)}} = \frac{\beta E(t)}{\mu^k n}. \quad (6)$$

Expression $\beta/\mu^k = (\alpha_k^{(i)} - 1)/\alpha_k^{(i)}$ is a positive fraction and equals the price-cost margin of a monopolist who manufactures product k . For any given level of consumption expenditure, $\Pi_k^{(i)}$ declines with k , as technological opportunities within a product generation decline.

2.3. R&D investment

New products are discovered through sequential and costly R&D races. R&D races are modeled as ‘invention lotteries’ with firm-specific risk but no aggregate risk. Denote with R_j the aggregate R&D resources devoted to the race, and with $\tau^{(i)}$ the time interval in sector i since the last breakthrough in that particular sector. Let $h(R_j)$ denote the deterministic duration of race j , if that race aims at discovering a product improvement. Let $h_0(R_j, \tau^{(i)})$ denote the duration of race j , if that race aims at discovering a breakthrough in sector i . The following assumptions specify the behavior of firms during a typical R&D race j , and describe the properties of the two duration functions.

- (A1) Once in an R&D race, each firm is required to commit a fixed amount of resources until the end of the race.
- (A2) The probability that firm l wins race j is given by R_l/R_j where R_l is the amount of R&D resources firm l devotes to race j .
- (A3) Exogenous innovation is allowed in a sense that $h(0) \geq 0$; $h_0(0, \tau^{(i)}) \geq 0$.
- (A4) The duration of race j decreases in its argument(s): $\partial h/\partial R_j < 0$, $\partial h_0/\partial R_j < 0$, $\partial h_0/\partial \tau^{(i)} < 0$.
- (A5) The cumulative R&D costs of each race, $C(\cdot)$ or $C_0(\cdot)$, increase in aggregate R&D resources devoted to the race: $\partial C(\cdot)/\partial R_j > 0$; $\partial C_0(\cdot)/\partial R_j > 0$, where $C(R_j) = R_j(e^{\rho h} - 1)$ and $C_0(R_j) = R_j(e^{\rho h_0} - 1)$.
- (A6) The duration of each race for a breakthrough is longer than the duration of each race for an improvement if both functions are evaluated at the same level of R&D: $h_0(R, \infty) > h(R)$.

Assumptions (A1) through (A5) are identical to those proposed by Segerstrom et al. (1990). Assumption (A2) makes the calculations tractable and allows us to

focus on how R&D investment changes across races. Although assumption (A3) allows exogenous technological discoveries, this paper focuses on equilibria with positive R&D investment. Assumption (A4) has two economic implications. First, it states that more R&D resources result in faster discoveries. Second, it captures the idea that the costs of a breakthrough in sector i depend on the exogenously accumulated basic research and scientific knowledge. Thus the longer is the time since the last breakthrough in a particular sector the lower are the costs of the next breakthrough. In a dynamic stable pattern of innovation, the role of $\tau^{(i)}$ is to provide a chronological ranking of innovations across sectors. If $h_0(\cdot)$ does not depend on $\tau^{(i)}$, then the intersectoral pattern of innovation is indeterminate.⁵ Finally, assumption (A6) implies that the discounted costs of a race for a breakthrough are higher than those of a race for any improvement. The following two specific functions satisfy assumptions (A1)–(A7): $h(R) = \ln[\gamma + (\delta/(1 + R))]/\rho$; and $h_0(R, \tau) = \ln[\gamma_0 + e^{-\tau} + (\delta/(1 + R))]/\rho$, where $\gamma_0 > \gamma \gg 1$ and $\delta > 0$ are parameters.

We assume that R&D races take place sequentially, and that there is only one race at any instant in time. One interpretation of this assumption is to consider an economy-wide R&D sector which develops new products for different sectors based on profitability considerations. Firms enter the most profitable R&D race until the expected discounted profits from winning the race are equal to their respective discounted R&D costs. When the most profitable R&D race breaks even, all other potential R&D races make losses if they are evaluated at the equilibrium R&D investment level. Because the duration function $h(\cdot)$ is identical across all races for improvements, it is obvious that the R&D sector cannot target improvement $k + 1$ until improvement k has been exhausted in all sectors.

The aggregate cumulative R&D costs of the j th race which targets the k th version of an arbitrary product generation are

$$C_k(R_j) = R_j r_c(t_j + h_k) \int_0^{h_k} \frac{1}{r_c(t_j + y)} dy, \quad (7)$$

where t_j denotes the time at which the j th race begins and $h_k(R_j)$ denotes the duration of the j th race. According to (7) the cumulative R&D costs are evaluated at the end of the R&D race. If the j th race targets a breakthrough, then $k = 0$ and the LHS in (7) becomes $C_0(R_j; \tau^{(i)})$.

⁵The dependence of $h_0(\cdot)$ on $\tau^{(i)}$ captures the ‘invention push’ hypothesis of innovation. See Gort and Wall (1986) for an excellent summary of the debate between the proponents and opponents of this invention push hypothesis.

The discounted profits of the j th race, which targets the k th version of a product generation, also evaluated at the end of the race, are

$$B_k(R_j) = \int_0^T \frac{r_c(t_j + h_k)}{r_c(t_j + h_k + y)} \Pi(t_j + h_k + y) dy. \quad (8)$$

Free entry into each R&D race ensures that $B_k(R_j) = C_k(R_j)$ for $k = 0, 1, 2, \dots$

2.4. The labor market

At each instant in time, workers can be employed in the R&D sector or in manufacturing of final goods. The latter consists of sectors with a monopolist and sectors where production of goods is characterized by perfect competition. Denote with m_j the number of sectors with monopolists during the j th R&D race. The full employment of labor condition becomes

$$L = R(t) + \left\{ \frac{E(t)}{n} (n - m_j) + \frac{E(t)}{n} \sum_{i \in M_j} \frac{1}{\alpha_i} \right\}. \quad (9)$$

The LHS of (9) is the economy-wide endowment of labor, and corresponds to the supply of labor. The first term of the RHS is the demand for labor in the R&D sector which equals $R(t)$ because one worker produces one unit of R&D services. The term in brackets captures the demand for labor in manufacturing. The first term in the bracket equals the output of all competitive sectors. Each sector produces $E(t)/n$ units of output because the price equals unity, and there are $(n - m_j)$ competitive sectors in the economy. The second term in the bracket equals the aggregate output produced by existing monopolists. The monopolist in sector i changes a price equal to $\alpha^{(i)}$, and therefore the quantity demanded is $E(t)/n\alpha^{(i)}$. The set M_j indicates the product versions produced by the existing monopolists during the j th R&D race. For example, if there are four monopolists during the j th race, two of which have discovered breakthroughs, one with a first improvement and another one with a second improvement, then $m_j = 4$ and $M_j = \{0, 0, 1, 2\}$.

Assumption (A1) requires $R(t) = R_j$ to remain constant during each race. Because there is only one race at each instant in time, the number of monopolists, m_j , cannot increase during a typical R&D race. However, it is possible for m_j and M_j to change during an R&D race because the patent length of an existing monopolist can expire. Without loss of generality, the following assumption simplifies the analysis and exposition of the next section:

(A7) The composition of existing monopolists, M_j , remains the same for the duration of a typical R&D race j .

Cheng and Dinopoulos (1992a) analyzed the effects of firm death during an R&D race, which basically results in upward jumps in aggregate expenditure $E(t)$ and the cumulative interest factor $r_c(t)$. These jumps do not change the qualitative results of the analysis.

3. Dynamic patterns of innovation and Schumpeterian growth

The results of the paper are summarized in three propositions. The first proposition establishes a basic property of the innovation pattern. The second proposition relates the existence of cyclical equilibria to the degree of diminishing returns to quality improvements. The third proposition states several comparative statics results. The Appendix contains the proofs of these propositions.

A dynamic general equilibrium is defined as the time paths of the cumulative interest factor, consumption expenditure, R&D investment, and commodity prices such that the demand for labor, for savings, and for every commodity is equal to its perspective supply at every instant in time. Given the choice of labor as the numeraire, the prices of all products produced under perfect competition are equal to unity, whereas each monopolist's price is given by the appropriate value of α_k .

Proposition 1. Suppose that the breakthrough duration function $h_0(R, \tau)$ is identical for all sectors and product generations. Then after the first round of breakthroughs in all sectors, all subsequent rounds of breakthroughs must follow exactly the same chronological order.

Proposition 1 allows us to focus on stable symmetric patterns of innovations, in which the composition of monopolists remains identical across cycles. We analyze two alternative cyclical patterns of innovation. In the first pattern, breakthroughs in different industries are separated by improvements. In the second pattern, breakthroughs in different sectors are clustered together.

Substituting α_k from (2) into the full employment condition (9) we obtain

$$L = R_j + \frac{E_j}{n} \left(n - \beta \sum_{q \in M_j} \mu^{-q} \right). \quad (10)$$

Assumptions (A1) and (A7) imply that R_j and the term in parenthesis on the RHS of (10) are constant during the j th R&D race. This means that aggregate expenditure $E(t) = E_j$ must remain constant during each R&D race. Eq. (5) implies that the cumulative interest factor equals $r_c(t) = e^{\rho t}$, and that the instantaneous interest rate $r(t)$ equals the constant subjective discount rate ρ . Of

course, because M_j changes across races, R_j and E_j change as well from one race to the next. These implications simplify expression (7) to

$$C_k(R_j) = [R_j(e^{\rho h_k(R_j)} - 1)]/\rho.$$

Performing the integration in (8), using (6), and substituting E_j from the full employment of labor condition, we get

$$B_k(R_j) = \frac{\beta(1 - e^{-\rho T})(L - R_j)}{\rho \mu^k \left(n - \beta \sum_{q \in M_j} \mu^{-q} \right)}.$$

Free entry into R&D race j which targets innovation k implies that cumulative costs $C_k(R_j)$ equal the discounted benefits $B_k(R_j)$:

$$R_j(e^{\rho h_k(R_j)} - 1) = \frac{\beta(1 - e^{-\rho T})(L - R_j)}{\mu^k \left(n - \beta \sum_{q \in M_j} \mu^{-q} \right)}. \quad (11)$$

Eq. (11) determines the unique value of R_j for each R&D race given our assumptions. The LHS of (11) is the cumulative R&D costs of an arbitrary race times ρ . It increases in R_j because of assumption (A5). The RHS of (11) is proportional to the discounted benefits of R&D. It is greater than zero for $R_j = 0$, becomes zero when $R_j = L > 0$, and decreases linearly in R_j . Thus, there exists a unique $R_j > 0$ which solves Eq. (11).

Proposition 2. There exists a value of the degree of diminishing returns to technological improvements $\bar{\mu}$ such that: (i) for $\mu \in (1, \bar{\mu})$ cyclical stable equilibria consisting of breakthroughs and improvements are the only possible ones; (ii) for $\mu \in (\bar{\mu}, \infty)$ there exists a unique steady-state equilibrium consisting of breakthroughs only.

Proposition 2 states that for a moderate degree of declining technological opportunities, a steady-state equilibrium does not exist. The critical value $\bar{\mu}$ depends on all parameters of the model. The economic intuition of Proposition 2 is straightforward. By definition, the benefits of a breakthrough exceed those of each subsequent improvement, but assumption (A6) implies that the costs of a breakthrough exceed those of any improvement as well. At the end of each race, the economy-wide R&D sector faces the decision of whether to target an improvement or a breakthrough. If μ is small, then the profitability of the first

improvement is higher than that of the cheapest breakthrough. Thus, the economy targets an improvement instead of another breakthrough. Of course, diminishing returns to technological improvements imply that a breakthrough will be targeted eventually. Any pattern of innovation which consists of breakthroughs and improvements implies a corresponding sequence of distinct levels of R&D investment, R_j , and aggregate expenditure, E_j .

Large values of μ imply that the first and all subsequent improvements are too expensive relative to the most profitable breakthrough at each instant in time. Thus, a steady-state equilibrium that consists only of breakthroughs is the only possible one. Eq. (11) determines the steady-state value of R_j for all j by substituting $h_k(R_j)$ with $h_0(R, \tau)$ and by setting $\mu = 1$. The number of sustainable monopolists is given by $m = T/h_0(R, \tau)$ and $\tau = (n - 1)h_0(R, \tau)$. This equilibrium was analyzed by Segerstrom et al. (1990) in a North–South model of trade and Schumpeterian growth. We therefore concentrate on describing two symmetric patterns of cyclical Schumpeterian growth.

The first pattern of innovation generates an intrasectoral cycle which moves across sectors over time. Suppose that the economy consists of two sectors and each cycle contains a breakthrough and two improvements. The chronological sequence of innovations which correspond to the intrasectoral cycle is

$$\{x_0^{(1)}, x_1^{(1)}, x_2^{(1)}, x_0^{(2)}, x_1^{(2)}, x_2^{(2)}, x_0^{(1)}, x_1^{(1)}, x_2^{(1)}, \dots\}.$$

In other words, the intrasectoral innovation pattern $\{x_0, x_1, x_2\}$ shifts across the two sectors. The second pattern of innovation generates an intersectoral cycle in which all sectors experience a breakthrough, then the first improvement and so on. The chronological sequence of innovations associated with the intersectoral pattern in the example is

$$\{x_0^{(1)}, x_0^{(2)}, x_1^{(1)}, x_1^{(2)}, x_2^{(1)}, x_2^{(2)}, x_0^{(1)}, x_0^{(2)}, x_1^{(1)}, x_1^{(2)}, x_2^{(1)}, x_2^{(2)}, \dots\}.$$

The cycle of innovations consists of the sequence $\{x_0^{(1)}, x_0^{(2)}, x_1^{(1)}, x_1^{(2)}, x_2^{(1)}, x_2^{(2)}\}$ which repeats itself over time indefinitely. A sector starts with a breakthrough which is diffused across the other sector sequentially through learning-by-investing. After the breakthrough has been adopted by both sectors, the first industry initiates improvement x_1 which is diffused sequentially and so on.

Consider the determination of R&D investment and aggregate consumption expenditure over a typical intrasectoral cycle. The horizontal axis of Fig. 1 measures the R&D investment level, R . The upper vertical axis measures the discounted benefits $B_k(R)$ and the cumulative R&D costs of a race $C_k(R)$. The lower vertical axis measures the aggregate consumption expenditure as a function of R&D, $E(R)$. The two upward-sloping curves starting at the origin denote the cumulative R&D costs of a breakthrough for a given equilibrium value of $\tau^{(i)} = \bar{\tau}$ which is $C_0(R, \bar{\tau})$, and the cumulative costs of all improvements, $C(R)$.

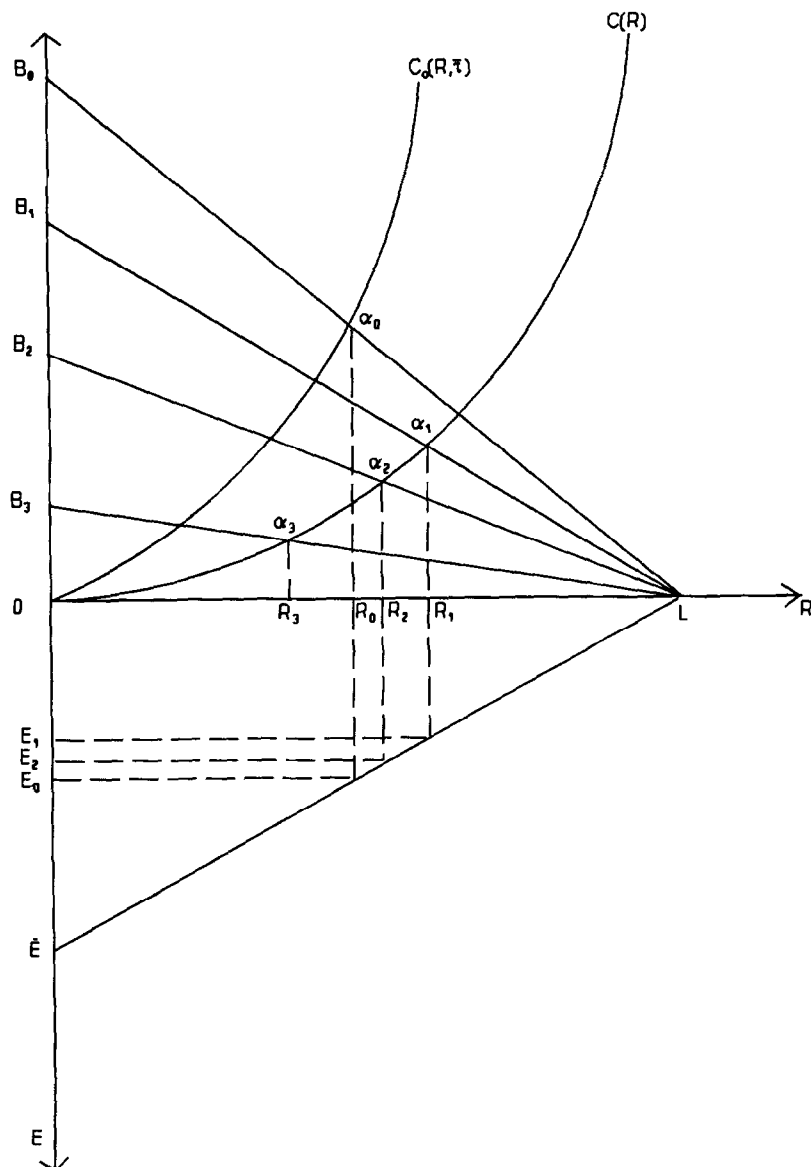


Fig. 1. Intrasectoral cyclical pattern of innovation.

Each negatively sloped line is the graph of the RHS of Eq. (11) for different values of k , and corresponds to the discounted profits of an R&D race for a particular innovation. These lines rotate counter-clockwise around point L as

the profitability of innovation decreases in k . Fig. 1 illustrates four lines with line B_0L associated with a breakthrough and the other three with the first three improvements. The negatively sloped line $L\bar{E}$ in the lower panel corresponds to the full employment of labor condition (10) and captures the negative relationship between labor allocated to R&D and labor allocated to manufacturing of final goods.

In order to simplify the exposition and geometry, we concentrate on an intrasectoral cycle consisting of a breakthrough and two improvements. The cycle has to start with a breakthrough in a particular sector, say sector 1. The amount of labor devoted to the race for a breakthrough is given by the intersection between $C_0(R, \bar{\tau})$ and B_0L at point α_0 . This intersection determines the unique values of R&D investment, R_0 , and consumption expenditure, E_0 . When the race for a breakthrough ends, the economy can target the first improvement or another breakthrough in another sector (due to Proposition 1). The next R&D race targets improvement α_1 . The equilibrium R&D investment is given by the intersection of $C(R)$ and line B_1L at point α_1 , which determines R&D investment, R_1 , and consumption expenditure, E_1 . The reason that the first improvement is targeted instead of another breakthrough is the following: At the equilibrium level of R&D for the first improvement, R_1 , all breakthroughs make negative discounted profits because $C_0(R_1, \bar{\tau}) > B_0(R_1)$, and $C_0(R, \bar{\tau})$ is the cumulative cost of the *most* profitable breakthrough. After the first improvement occurs, the R&D sector can target the second improvement or a breakthrough in another sector. In Fig. 1 we have $C(R_2) = B_2(R_2)$ and $C_0(R_2, \bar{\tau}) > B_0(R_2)$ and consequently the second improvement is targeted and results in R_2 and E_2 . However, by the time the fourth race starts, sector 2's cumulative cost has declined to $C_0(R, \bar{\tau})$ making a breakthrough in sector 2 more profitable than the third improvement in sector 1 because $B_3(R_3) = C(R_3)$, but $B_0(R_3) > C_0(R_3, \bar{\tau})$. Thus the sequences $\{R_0, R_1, R_2\}$ and $\{E_0, E_1, E_2\}$ repeat themselves forever moving across sectors starting with sector 1. This pattern of innovation exhibits a negative duration dependence within a sector and a negative correlation across sectors at any point in time.

Fig. 2 illustrates an intersectoral pattern of cyclical innovation consisting of a breakthrough and the first improvement. The figure reproduces the basic features of Fig. 1. It is assumed that breakthroughs are clustered and that the maximum gap between two adjacent breakthroughs equals $\tau'' - \tau' > 0$. In other words, the cumulative costs curves of all clustered breakthroughs are located above curve $C_0(R, \tau')$ and below curve $C_0(R, \tau'')$. The cycle starts with the cumulative costs of all breakthroughs that are located to the right of curve $C_0(R, \tau')$. Proposition 1 implies that each of the n cumulative cost curves of breakthroughs is distinct. Thus all breakthroughs dominate the first improvement, because all possible improvements make losses when evaluated at $R \geq R_0$. After n breakthroughs, if the maximum gap $\tau'' - \tau'$ is long, R'_0 is located to the left of R_1 . The first improvement dominates all breakthroughs and will be

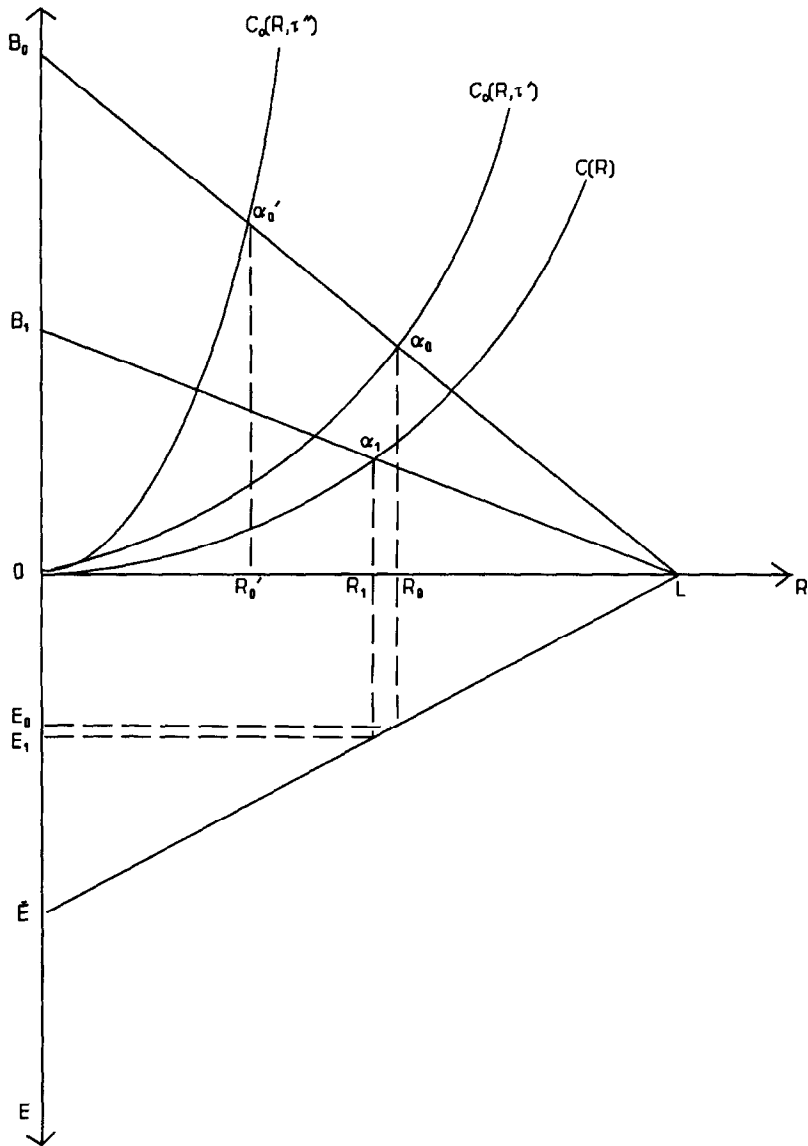


Fig. 2. Intersectoral cyclical pattern of innovation.

pursued in all sectors. During the intersectoral round of the first improvement, curve $C_0(R, \tau'')$ shifts to the right and R'_0 approaches R_1 . This means that the cluster of breakthroughs dominates the second improvement which will not be pursued. An intersectoral pattern of endogenous cycles and growth emerges in

which a round of breakthroughs is followed by a round of first improvements in all sectors. If the maximum gap between two adjacent breakthroughs is large enough, more than one round of improvements occurs. Innovation exhibits a negative correlation across sectors at any point in time and a negative dependence within a sector.

Both patterns of cyclical innovation result in economy-wide cycles of R&D investment, consumption expenditure, the price-cost margin, and the cumulative interest factor. For any given number of innovations, the intersectoral cycles are longer than the intrasectoral cycles. If the cumulative costs of breakthroughs do not depend on τ , then the intersectoral pattern of cyclical innovation cannot occur.

Cheng and Dinopoulos (1992a) analyzed the comovements of the model's key macroeconomic variables. In what follows we explore the relationship between the duration of a typical endogenous cycle and the long-run Schumpeterian growth trend. The instantaneous utility is a step function of time and jumps upward every time a new product is discovered. The length (duration) of each step equals to the duration of the relevant R&D race. The average long-run growth is given by

$$G = \left(\sum_{q=0}^s \ln x_q \right) / \left(h_0(R_0, \tau) + \sum_{q=1}^s h(R_q) \right), \quad (12)$$

where s is the number of technological improvements in a typical cycle. Because changes in aggregate expenditure are transitory, the numerator of (12) equals the net increase in the logarithm of instantaneous utility over a cycle. The denominator equals the duration of a cycle which decreases in R&D investment. For any given level of technological opportunities, (12) implies that the long-run growth trend is negatively related to the duration of a typical cycle. The following proposition summarizes the comparative statics results of the model.

Proposition 3. Suppose that the composition of monopolists remains invariant to marginal changes in the parameters of the model. The long-run growth trend increases and the duration of a typical cycle decreases: with an increase in the labor endowment, L , with an increase in the level of technological opportunities, β , with an increase in the patent length, T , and with a reduction in the degree of diminishing returns to technological improvements, μ .

Proposition 3 establishes specific links between the duration of cycles and the long-run growth trend. For small changes in the parameters of the model higher long-run growth is negatively related to the duration of cycles. However, large changes in the parameters could change the composition of each cycle in an ambiguous way by making more improvements profitable.

4. Concluding remarks

This paper analyzed two cyclical patterns of innovation which resulted in endogenous cycles and Schumpeterian growth within a multisectoral general equilibrium model. Cheng and Dinopoulos (1992a) have analyzed the comovements of macroeconomic variables implied by the two cyclical patterns. R&D investment, the value of assets, and the average price-cost markup of the economy move together in a procyclical fashion, if business cycles are defined as real output fluctuations around its growth trend. Cheng and Dinopoulos (1993) have generated stochastic cyclical patterns of growth in a single-sector general equilibrium model.

One should be careful in interpreting the cycles generated by the model as economic recessions and booms. The model abstracts from unemployment considerations, capital accumulation, macroeconomic policies, and international interactions which play a crucial role in the causation of economic fluctuations. Incorporating some of the above-mentioned elements into the model is a task that deserves further research.

Appendix

A.1. Consumer's problem

The demand for any state-of-the-art quality product produced competitively is given by E/n , where $E(t)$ is the aggregate consumption expenditure. The demand for each product produced by a monopolist is given by $E/(n\alpha_k^{(i)})$, where $\alpha_k^{(i)}$ is the monopoly price. Substituting these demands into (3), one obtains the indirect utility function which is given by

$$V = (E/n)^n \Gamma, \quad (\text{A.1})$$

where $\Gamma = \prod_{i=1}^n \prod_{s=1}^{k(i)} \alpha_s^{(i)} / \prod_{i \in M} \alpha^{(i)}$, $k(i)$ is the state of the art product in industry i , and M is the set of industries whose newest products are currently produced by dominant firms. Obviously Γ is a function of time and depends on the pattern of innovation as well as the imitation lag, but is regarded as a parameter by an individual consumer. The consumer chooses E – subject to her wealth constraint to maximize discounted utility. Recognizing that the logarithmic function serves to decouple E and Γ in (A.1) and that both n and Γ are regarded as parameters by the consumer, the problem is equivalent to

$$\max_{\{E(t)\}} \int_0^\infty e^{-\rho t} \ln E(t) dt \quad \text{s.t.} \quad A(t) = \int_0^t \frac{r_c(t)}{r_c(s)} [L - E(s)] ds \geq 0, \quad (\text{A.2})$$

where $A(t)$ denotes the consumer's financial wealth at t , $r_c(t)$ is the cumulative interest factor for t , and L is both the supply of labor as well as labor income. The solution to (A.2) is given by Eq. (5).

A.2. Proof of Proposition 1

At the beginning of time, there are no breakthroughs so that $\tau^{(i)} = \infty$ for all i . The production technology of products $x_0^{(i)}$, $i = 1, \dots, n$, is common knowledge. By indexing the sectors appropriately, the first R&D race aims at a breakthrough in sector 1. At the beginning of the second race, the value of τ for industry 1, $\tau^{(1)}$, equals zero. Thus, the second R&D race is either for a breakthrough in another sector, indexed by 2, or for the first improvement in sector 1. The chosen innovation must 'dominate' all other potential innovations in the sense that at its break-even R&D level, R , all other potential R&D races make losses or at best they break even. At the end of a race for a breakthrough in sector 2, $\tau^{(1)} > \tau^{(2)} = 0$. More generally, whatever the intervening innovations, at the end of the race for the first breakthrough in sector n , we have $\tau^{(1)} > \tau^{(2)} > \dots > \tau^{(n)} = 0$. This inequality implies that even though the chronological order of the first round of breakthroughs across sectors is arbitrary, all subsequent rounds must take place in exactly the same order as the first. Q.E.D.

A.3. Proof of Proposition 2

The unique symmetric steady-state equilibrium is defined by the following equations:

$$R_0(e^{\rho h_0(R_0, \tau_0)} - 1) = \frac{\beta(1 - e^{-\rho T})(L - R_0)}{(n - \beta m_0)}, \quad (\text{A.3})$$

$$\tau_0 = (n - 1)h_0(R_0, \tau_0), \quad (\text{A.4})$$

$$m_0 = T/h_0(R_0, \tau_0). \quad (\text{A.5})$$

Eq. (A.3) is derived from (11) by setting $k = 0$ and $\mu = 1$. Eq. (A.4) defines the maximum period between races for breakthroughs in each of the n sectors. Eq. (A.5) defines the steady-state number of sustainable monopolists. In the symmetric steady state defined by (A.3), (A.4), and (A.5), R&D investment, R , and all relevant endogenous variables are constant over time.

Fig. 3 illustrates the determination of R_0 by plotting the LHS and the RHS of Eq. (A.3) evaluated at the equilibrium levels of m_0 and τ_0 . The LHS of Eq. (A.3) is

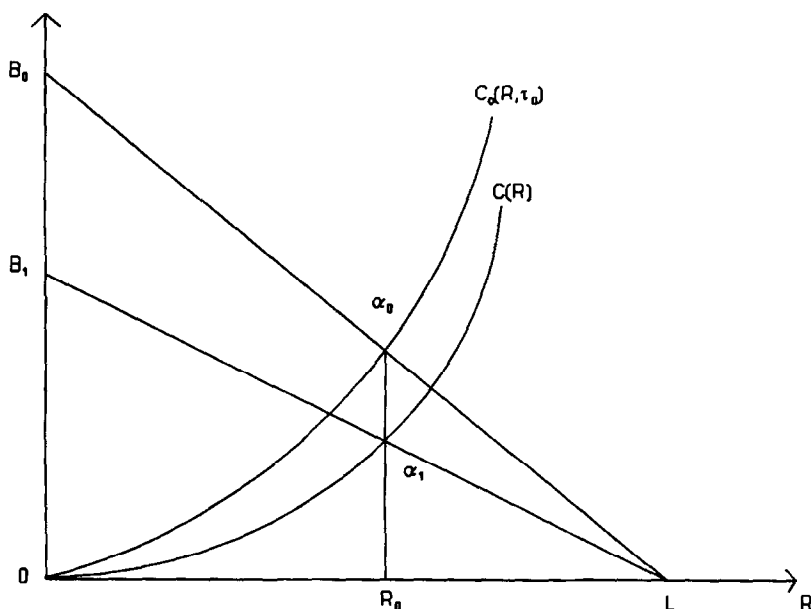


Fig. 3. Steady-state and cyclical patterns of innovation.

an upward-sloping curve, OC_0 , starting at the origin. The particular position of OC_0 captures the cumulative R&D costs of the sector with the longest time gap, which equals τ_0 . The RHS of Eq. (A.3) is a downward-sloping line, LB_0 , with positive intercepts in both axes. The unique intersection of curves OC_0 and LB_0 determines the value of R_0 .

Consider the corresponding equilibrium for the first improvement which is given by Eq. (11). Notice that (A.3) and consequently R_0 do not depend on the degree of diminishing returns, μ . The LHS of Eq. (11) is independent of μ as well and defines an upward-sloping curve, OC , which starts at the origin and lies below curve OC_0 . Parameter μ affects the RHS of Eq. (11). As μ approaches unity, the downward-sloping line approaches LB_0 . As μ increases, the LB_1 line rotates counterclockwise. Thus, there exists a value of $\mu = \bar{\mu} > 1$ such that the intersection of OC and LB_1 occurs at point α_1 which is located exactly below point α_0 .

If $\mu = \bar{\mu}$, then the most profitable breakthrough and the first improvement just break even and the economy is indifferent between the two. However, for $\mu \in (1, \bar{\mu})$, point α_1 is located to the right of point α_0 in Fig. 3, and the first improvement dominates the most profitable breakthrough. In this case, a steady-state equilibrium does not exist. Finally, for $\mu \in (\bar{\mu}, \infty)$, point α_1 is located to the left of point α_0 . The first (and all subsequent) improvements are

dominated by the most profitable breakthrough, and there exists a symmetric steady-state equilibrium consisting of breakthroughs only. Q.E.D.

A.4. Proof of Proposition 3

Differentiation of Eq. (11) with respect to the parameters of interest, and use of Eq. (12) establishes the comparative statics results of the proposition. Q.E.D.

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